28/05/2026

Assignment - 1

1. What is the output of the following and Why

S = " P y t h o n " (indices 0 1 2 3 4 5)
print (S[0], S[-1], S[6])

Ans :- Error → The Index number of S[6] is not available in "python" character

2. What is the difference between positive and Negative Indexing ? Given S = "Hello" what does S[-2] return?

Ans Different Between positive and negative indexing

Positive → It starts from left to right and it will start with index number 0

Negative Indexing → It start from right to left and it will start with Index number -1

S = " H e l l o " (indices 0 1 2 3 4; -5 -4 -3 -2 -1)
S[-2]

→ "l"

Page
Date

3.

Ans S = " code "
print (S [10])
print (S [1:10])

$\overset{0}{c}\,\overset{1}{o}\,\overset{2}{d}\,\overset{3}{e}$

In comment print (S [10]) there is No Index number 10 is available so it will be error

In Comment print (S [1:10]) there is No Index number 10 available But Index number 1, 2, 3, is available So the output will be, ode

4.

Ans S = " Mississippi "
Find index of first "S" and last "S"

$S = " \overset{0}{M}\ \overset{1}{I}\ \overset{2}{S}\ \overset{3}{S}\ \overset{4}{I}\ \overset{5}{S}\ \overset{6}{S}\ \overset{7}{I}\ \overset{8}{P}\ \overset{9}{P}\ \overset{10}{I} "$

First S → S. index ("S") → It will check from left side
2

Last S → S. RIndex ("S") → It will check from right side
6

Page
Date

5.

Ans s = "abcdef"

	a	b	c	d	e	f
	0	1	2	3	4	5
	-6	-5	-4	-3	-2	-1

print (s[-1], s[-3], s[-6])

output → F d a

6.

Ans s [start : stop : step]

start → The starting index numbe including the start index

stop → We need to provide the next index number upto which we need to print or to process

step → used for step increment and step decrement

7.

Ans s = "Hello World"

(index 0–9 written above the letters)

print (s[0:5]) → Hello

Print (s[5:7]) → World

Print (s[:5]) → Hell

Print (s[:7]) → Helloword

8.

Ans s = "Python"

print (s[4:1])

No output will occur due to greater Index number in the start

9.

Ans eg:: s = "python"

print (s[-3:]) → It will print last three characters

10.

Ans s = "abcdefgh"

print (s[::1]) ⟶ abcdefgh

print (s[::2]) ⟶ a c e g

print (s[1::2]) ⟶ b d f h

11.

Ans s = "python"

s[::-1] → To reverse the value

~~print(s[::-1])~~

-1 step value is used to reverse.

12.

Ans

```
        0 1 2 3 4 5 6 7
s = " a b c d e f g h"

print(s[::-1]) -> hgfedcba

print(s[::-2]) -> hfdb

print(s[6:1:-1]) -> gfedc
```

13.

Ans

```
      0 1 2 3 4 5 6 7 8 9
s = " a 1 b 2 c 3 d 4 e 5"

print(s[1::2])
```

14.

Ans

```
       0 1 2 3 4 5
s = " p y t h o n"

print(s[5:0:-1]) -> nohty

print(s[5::-1]) -> nohtyp

print(s[:0:-1]) -> nohty
```

15.

Ans

```
s = "hello"
print(s[100::-1]) -> olleh
print(s[::-100]) -> o
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Date

16.
Ans s = "abcdeFghiJ"

~~print~~ s[2:-1]

output → cdeFghI

17.
Ans S = "racecar"

S[::-1]

It is palindrome (After reversing the output is as same as input it is palindrom)

18.
Ans S= "Hello"
print(s[1:100]) → ello
print(s[-100:3]) → Hel
print(s[100:200]) → No out will be printed

19.
Ans S.replace("world", "python")

20.
Ans S= "abcdeFghiJ" (indices 0 1 2 3 4 5 6 7 8 9 written above the letters)
My_slice = slice(1, 8, 2)
print(s[my_slice])

Output → b d F h